Digit presentation of the prime-field Semaev system: a native finite degree ladder with independent primes

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Track: algebraic (a degree study; not a cost). Nothing here is an ECDLP cost, so there is no dashboard point and no `benchmarks/` entry. Lead: `LEAD-ICP-digit-presentation-solving-degree`.

Result. The last fall degree `d_last` of the base-2 digit-presented Semaev system grows with the digit count `s`, at the closed-form rate, on every independent draw. The lane is **exhausted at the measured scale** (`m = 2`, `s ≤ 7` unsatisfiable and `s ≤ 8` planted; `m = 3`, `s = 4`). It does not reopen.

- **`d_last = 4 + ⌊s/2⌋` on 106 of 106 real draws at `m = 2`** (60 unsatisfiable at `s = 2..7`, 46 planted at `s = 2..8`): 5, 5, 6, 6, 7, 7, 8. In every one, `d_last` equals the first fall degree `ffd`. Each draw has its own prime, curve and target.
- **`m = 3`:** `ffd = d_last = 13` at `s = 4` (4 draws, closure run); `ffd = 13` at `s = 4, 5` (first fall only, 4 draws each).
- **Support-matched null:** `ffd = d_last = s + 3` on 44 of 44 draws. The real system falls strictly earlier from `s = 3` (the gap is 0, 1, 1, 2, 2, 3 at `s = 2..7`).
- **Block-factored null** (∏ `q_k`, random forms on the support of `ℓ_k^e`): it reproduces the real `ffd` and `d_last` on **44 of 44 draws**. So the early fall below the support-matched null is explained by the block structure alone. It carries nothing curve-specific.
- **Plain presentation** (`F_p[x_1, x_2]/(x_i^p − x_i)`, the same engine): `d_last = ffd = p + 2` on 32 of 32 draws, `p = 7..31`. The plain degree is linear in `p`; the digit degree is linear in `log p`.
- **But the matrix is nearly the whole ring.** At `d_last` the degree-filtered Macaulay space spans 100%, 98%, 96%, 83%, 81% and 60% of the `2^n` monomials at `s = 2..7` (§3, last column), and 60% at `s = 8`. A low degree does not make a small system at this scale.

Public synthetic curves; no security claim. Registered in `PROTOCOL.md` before the counted runs, with one amendment (`AMENDMENT-1.md`).

Labels. `[measured]`: native `digit_fall` records in `results/`. `[derived]`: arithmetic on those records. `[predicted]`: the registered table. `[cited]`: the ledger's closed form, not proved here.

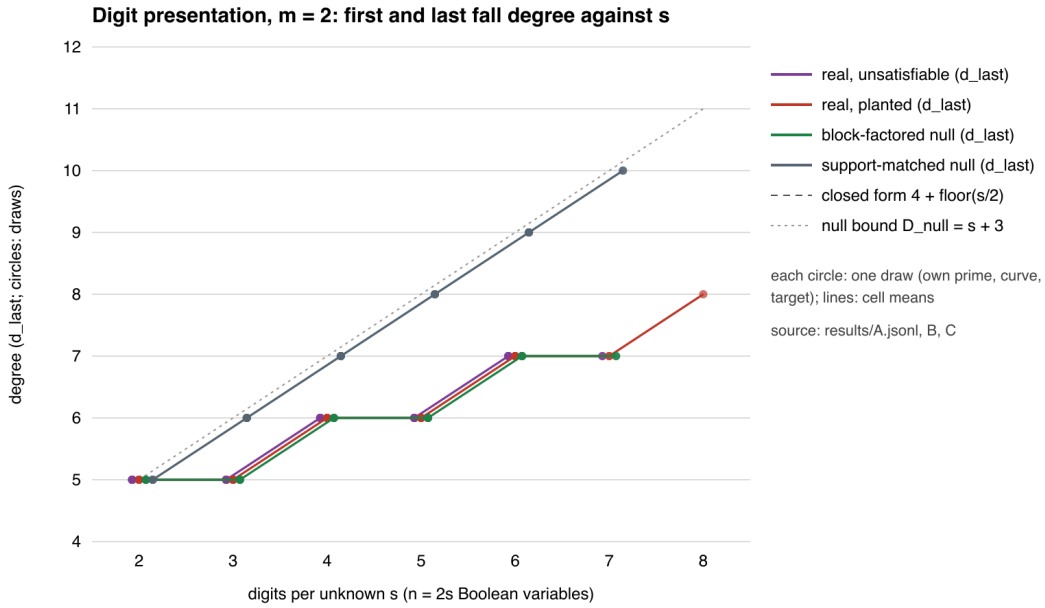


Figure 1: Last fall degree of the digit-presented system against s, m = 2: real unsatisfiable and planted draws, and the two nulls. Dashed: the closed form 4 + floor(s/2); dotted: the null bound s + 3. Source: degree_vs_s.svg.

1. What was run `[measured]`

The object is $\tilde{S}$ = `S_{m+1}(ℓ_1, …, ℓ_m, x_R)` reduced in `B = F_p[a]/(a² − a)`, with `ℓ_k = Σ_{i<s} 2^i a_{k,i}`, `n = m s`, over $y^2 = x^3 + ax + b$.

- **Engine.** `digit_fall.rs` is standard-library Rust. It ports the Huang–Kosters–Yeo engine `examples/hky_last_fall_degree.rs` from `F_2` to `F_p`. The conventions (Macaulay rows with the degree rule, mutant closure, `e0`, `d_last`, `ffd`) are listed in `PROTOCOL.md` §2.
- **Exactness.** `B` is a function algebra on `{0,1}^n`, so `(f) = I(S)` for the zero set `S`. `dim(I ∩ R_{≤c})` and `DG` come from the evaluation rank on `S`, with no Gröbner basis. A value is exact whenever its run is not marked censored. No run was censored.
- **Validation.** On seed 7 it first reproduced the ledger's integers: `ffd = d_last = 5, 5, 6` at `s = 2, 3, 4` and 13 at `m = 3, s = 4`. Those draws are not in the panel.
- **Independence.** Each draw is a new prime in `(2^b, 2^{b+1})`, `b = min(2s + 1, 11)`, a new curve and a new target. This fixes the pseudoreplication of EV-PFDR-99c699, where a handful of integers were repeated 120 times. The `primes` column shows how many distinct primes each cell used. At `s = 2` the interval holds only 7 primes, so 5 distinct ones appear in 12 draws; every draw still has its own curve and target.
- **Replication.** 238 draws: A (60), B (46), C (88), D (12), E (32). 0 censored, 0 errors. The 46 planted draws are panel B at `s = 2..8`.

2. Predictions [predicted] and what happened [measured]

registered prediction	draws holding	draws failing
P1 `ffd = 4 + ⌊s/2⌋`, m = 2, real draws	106 / 106	0
P1 `ffd = 13`, m = 3, s = 4, 5	12 / 12	0
P2 `d_last = ffd`, real unsatisfiable draws	60 / 60	0
P5 `d_last = ffd`, planted draws	46 / 46	0
P3 support-matched null: `ffd = d_last = s + 3`	44 / 44	0
P4 block-factored null reproduces the real values	44 / 44	0
P6 plain: `d_last = ffd` linear in `p`	32 / 32 (`= p + 2`)	0

Decision rule (PROTOCOL §5). Reopen if `d_last` at `s = 7` is at most `d_last` at `s = 3`, or if any real draw has `d_last` below the block-factored null or below `ffd`. None happens: `d_last(7) = 7 > d_last(3) = 5`, and the real system equals the block null on every draw. The lane is exhausted at the measured scale.

3. The ladder, cell by cell [measured]

`nsol` is the number of digit solutions on the grid. In every cell the draws are distinct (each has its own curve and target) and none is censored; `cols` is `Σ_{j ≤ d_last} C(n, j)` ([derived] from the measured `d_last`).

m	arm	s	draws	primes	nsol	closed form	ffd	d_last	D_null at	cols d_last	cols / 2^n
2	U	2	12	5	0	5	5	5	5	16	1.00
2	U	3	12	10	0	5	5	5	6	63	0.98
2	U	4	12	11	0	6	6	6	7	247	0.96
2	U	5	12	11	0	6	6	6	8	848	0.83
2	U	6	8	8	0	7	7	7	9	3302	0.81
2	U	7	4	4	0	7	7	7	10	9908	0.60
2	P	2	8	4	{1,2,3}	5	5	5	5	16	1.00
2	P	3	8	8	{1,2,3}	5	5	5	6	63	0.98
2	P	4	8	7	{2,4}	6	6	6	7	247	0.96
2	P	5	8	8	2	6	6	6	8	848	0.83
2	P	6	8	8	{2,4,6}	7	7	7	9	3302	0.81
2	P	7	4	4	{2,6,8,14}	7	7	7	10	9908	0.60
2	P	8	2	2	{12,20}	8	8	8	11	39203	0.60
2	Usupp	2	8	5	0	5	5	5	5	16	1.00
2	Usupp	3	8	6	0	5	6	6	6	64	1.00
2	Usupp	4	8	8	0	6	7	7	7	255	1.00
2	Usupp	5	8	8	0	6	8	8	8	1013	0.99
2	Usupp	6	8	8	0	7	9	9	9	4017	0.98
2	Usupp	7	4	4	0	7	10	10	10	15914	0.97
2	Ublock	2	8	5	0	5	5	5	5	16	1.00
2	Ublock	3	8	7	0	5	5	5	6	63	0.98
2	Ublock	4	8	8	0	6	6	6	7	247	0.96
2	Ublock	5	8	8	0	6	6	6	8	848	0.83
2	Ublock	6	8	8	0	7	7	7	9	3302	0.81
2	Ublock	7	4	4	0	7	7	7	10	9908	0.60
3	U	4	4	4	0	13	13	13	13	4096	1.00
3	U (first fall)	4	4	4	0	13	13	n/a	13	n/a	n/a
3	U (first fall)	5	4	4	0	13	13	n/a	14	n/a	n/a

Plain control (`F_p[x_1, x_2]/(x_i^p − x_i)`, 4 draws per `p`, each a new curve and target):

p	nsol	ffd	d_last	d_last – p	grid points p²
7	{4,6,8}	9	9	2	49
11	{9,10,14,15}	13	13	2	121
13	{10,11,17,19}	15	15	2	169
17	{13,16,18,22}	19	19	2	289
19	{12,17,21,23}	21	21	2	361
23	{16,24,25,26}	25	25	2	529
29	{22,28,29,38}	31	31	2	841
31	{26,28,30,36}	33	33	2	961

The plain control and the digit presentation cover different unknown ranges: plain covers all of `F_p`, digit covers `2^s ≤ p` values. Compare the growth, not the level.

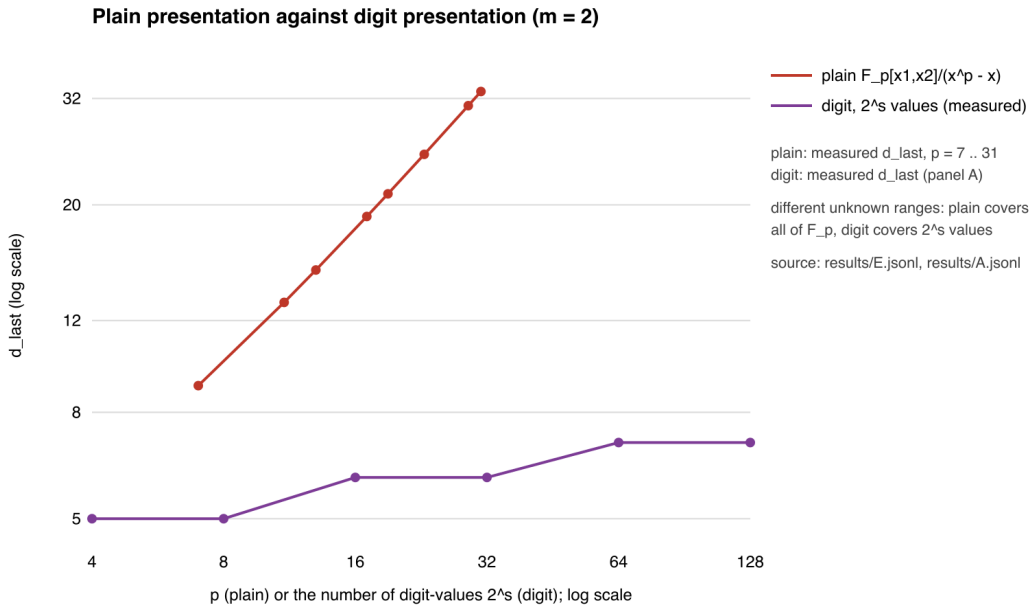


Figure 2: Last fall degree of the plain presentation (all of F_p, p = 7..31) and of the digit presentation (2^s values), log scale. Source: plain_vs_digit.svg.

4. Reading

1. **The digit presentation does not have a bounded last fall degree.** `d_last` rises by 1 every 2 digits, on 106 independent draws, in agreement with the ledger's closed form `d_ff = m 2^{m−1} + ⌊(s − 2^{m−1})/2⌋ + 1 [cited]`. That form is reproduced exactly for `d_last` here as well as for `ffd`.

2. **No elliptic early fall.** Block-factored random forms have the same first and last fall degree as the Semaev system on every draw. The real system differs from the support-matched null only through the block structure that ℓ_k^e already gives.
3. **Low degree, big matrix, at this scale.** d_last is far below the plain presentation's p + 2 (exponentially, in s). But the Macaulay space at d_last spans 60–100% of the ring. The fraction falls slowly (0.81 at s = 6, 0.60 at s = 7 and s = 8). [derived] Asymptotically d_last / n → 1/(2m), so the column count is about 2^{n·H(1/4)} at m = 2, which is not small; this is the ledger's σ > 1 point (D7 of H-PFDR-4148b8), not re-derived here.
4. **What this adds to the ledger.**
   - A native F_p engine that reproduces the Python integers.
   - Independent primes, curves and targets per draw.
   - The last fall degree to s = 7 (the ledger stopped at s = 5).
   - A planted (satisfiable) arm and m = 3 at s = 4.
   - The plain-presentation comparator.

Scale reached, and what would reopen it. Exact values for m = 2, s ≤ 7 (unsatisfiable) and s ≤ 8 (planted), m = 3, s = 4; first fall only at m = 3, s = 5. The lane reopens only if a larger s breaks the closed form (the next cell is s = 9, predicted d_last = 8; n = 18, up to about 2^18 columns), or if a different presentation (the adaptive staircase arm, H-PFDR-f3fe89, not run here) has a degree or matrix below the block null.

5. Limits

- **Toy scale.** p < 4096, s ≤ 7, m ≤ 3. The unsatisfiable arm stops at s = 7: for s ≥ 8, p < 4096 leaves no unsatisfiable target on the 2^16-point grid. The limit comes from the u32 accumulators (p < 2^12), not from mathematics.
- **s = 8.** Only the planted arm, 2 draws, n = 16: d_last = ffd = 8, as predicted, with 39,203 columns at c = 8 (about 11 minutes a draw). Two draws: a prediction check, not a distribution.
- **A degree is not a cost.** A fall degree is not a solving degree, a cost or an exponent. The column count is a size measure of the Macaulay space, not a priced operation count.
- **One convention.** The function-algebra quotient B (field equations free). The ordinary-ring convention of research/guarded_fall_degree_20261028/ is a different object, with proven bounds d_F^{ord} ≤ 3·ℓ + 2 for the bit-restricted presentation.
- **Not run.** The adaptive staircase arm (H-PFDR-f3fe89) and m = 3, s ≥ 5 last fall. These are the part of the lead's next action that remains open.
- **Pre-registration disclosure.** Validation draws (seed 7) preceded registration. Amendment 1 fixed a hang in the planted-target search; panel A (60 draws) was produced by the registered binary and the amended binary reproduces it line for line on a re-run of its s = 3 cell. A first version of the block null had no constant term (so q(0) = 0 made it always satisfiable and it never gave an unsatisfiable instance); it was fixed before any panel run.

- **Formalization.** No new theorem is claimed. `lean/NullTable.lean` (core Lean 4.34.1, `by decide`, no axioms for all six theorems) checks the registered table's arithmetic only. The closed form is `[cited]`; its proof needs Wilson's rank theorem on inclusion matrices (H-WIL), the graded annihilator of `(Σ a_i)^e` in the squarefree algebra and the tensor-kernel formula. None is in core Lean, so a faithful theorem is blocked on those three dependencies.

Suggested atlas change (not made here). Keep `LEAD-ICP-digit-presentation-solving-degree` as `open_next_step`, but record the finite ladder as `exhausted_at_scale` at `m = 2`, `s ≤ 8` and `m = 3`, `s = 4`; the open item is the adaptive staircase arm. Evidence: `research/prime_digit_presentation_20261029/`.

6. Files

- `PROTOCOL.md`, `REGISTRATION.sha256`, `BINARY.sha256`, `registered-at.txt`: the registration.
- `AMENDMENT-1.md`, `AMENDMENT-1-NOTE.md`, `REGISTRATION-AMENDMENT-1.sha256`, `BINARY-2.sha256`: the amendment and a post-run note (`run.sh` also changed; `REGISTRATION.sha256` reports `FAILED` for the two amended files, as expected).
- `digit_fall.rs`: the engine (`rustc -O digit_fall.rs`). `run.sh`: the panel. `derive.mjs`: the tables and figures (tooling).
- `results/{A..E}.jsonl`: one record per draw. `analysis/tables.md`, `analysis/summary.json`.
- `degree_vs_s.svg/png`, `plain_vs_digit.svg/png`, `report.pdf`.
- `lean/NullTable.lean`, `lean/lean-toolchain`.